

1 Problem 1

For this problem, I copied the output from my code into Google Gemini to make a nicely formatted LaTeX table, which is seen in Table 1.

Table 1: Side-by-Side Comprehensive LCG Period Analysis

K	Formula	N = 6, M = 64			N = 7, M = 128			N = 8, M = 256		
		a	p	p/M	a	p	p/M	a	p	p/M
3	$K + 0$	3	32	0.500000	3	64	0.500000	3	128	0.500000
	$K + 1$	4	3	0.046875	4	4	0.031250	4	4	0.015625
	$2K + 1$	7	16	0.250000	7	32	0.250000	7	64	0.250000
	$3K + 1$	10	6	0.093750	10	7	0.054688	10	8	0.031250
	$4K + 1$	13	64	1.000000	13	128	1.000000	13	256	1.000000
	$5K + 1$	16	2	0.031250	16	2	0.015625	16	2	0.007813
	$6K + 1$	19	32	0.500000	19	64	0.500000	19	128	0.500000
	$7K + 1$	22	6	0.093750	22	7	0.054688	22	8	0.031250
	$8K + 1$	25	64	1.000000	25	128	1.000000	25	256	1.000000
	$9K + 1$	28	3	0.046875	28	4	0.031250	28	4	0.015625
	$12K + 1$	37	64	1.000000	37	128	1.000000	37	256	1.000000
	$16K + 1$	49	64	1.000000	49	128	1.000000	49	256	1.000000
7	$K + 0$	7	16	0.250000	7	32	0.250000	7	64	0.250000
	$K + 1$	8	2	0.031250	8	3	0.023438	8	3	0.011719
	$2K + 1$	15	8	0.125000	15	16	0.125000	15	32	0.125000
	$3K + 1$	22	6	0.093750	22	7	0.054688	22	8	0.031250
	$4K + 1$	29	64	1.000000	29	128	1.000000	29	256	1.000000
	$5K + 1$	36	3	0.046875	36	4	0.031250	36	4	0.015625
	$6K + 1$	43	32	0.500000	43	64	0.500000	43	128	0.500000
	$7K + 1$	50	6	0.093750	50	7	0.054688	50	8	0.031250
	$8K + 1$	57	64	1.000000	57	128	1.000000	57	256	1.000000
	$9K + 1$	64	1	0.015625	64	2	0.015625	64	2	0.007813
	$12K + 1$	85	64	1.000000	85	128	1.000000	85	256	1.000000
	$16K + 1$	113	64	1.000000	113	128	1.000000	113	256	1.000000

From my results, when the multiplier, a , equal $4N + 1$, the linear congruential generator achieves a full period. In this table, a full period is denoted via bold text with a gray background. For every case of N and K , we observe a full period occurs when the period of the linear congruential generator equals $M = 2^N$. In this context, a full period means the period of the lcg equal $M = 2^N$.

In fact, from these results, we can make a stronger claim, which seems to hold until $a = 16K + 1$, that every multiplier of the form:

$$a = 4ZK + 1, \quad 1 \leq Z \in \mathbb{Z} \quad (1)$$

achieves a full period.

2 Problem 2

I again, formatted my results with Google Gemini.

Table 2: Comprehensive Quality, Period, and Ratio Analysis for Problem 2 ($M = 2^{24}$)

Case	a	b	x_0	Period (p)	p/M Ratio	χ^2 Score	Observed Moment	Moment Residual
(a)	1812433253	1	69069	2^{24}	1.0	1.43×10^{-6}	0.500000	2.98×10^{-8}
(b)	1812433253	69069	69069	2^{24}	1.0	1.43×10^{-6}	0.500000	2.98×10^{-8}
(c)	69069	69069	69069	2^{24}	1.0	1.43×10^{-6}	0.500000	2.98×10^{-8}
(d)	65539	0	69069	2^{22}	2^{-2}	5.72×10^{-6}	0.500001	1.19×10^{-7}
(e)	65539	0	1024	2^{12}	2^{-12}	0.005859	0.499878	1.22×10^{-4}
(f)	1024	0	69069	2^2	2^{-22}	11.000000	0.258064	0.241936

2.1 χ^2 Test

For the χ^2 analysis, I calculated determined the leading digit of each η in the whole period. Then, I counted the total number of times each leading digit occurred. Next, for the expected count of each leading digit, I divided the period by 10, one for each decimal digit, and assumed the same frequency of occurrence for all digits. For the χ^2 test, there are 9 degrees of freedom as $9 = 10$ possible digits - 1. With a significance level, α , of 0.05, we fail to reject the null hypothesis if $\chi^2 \leq 16.92$

- Cases a, b, and c have full periods and very small χ^2 values on the order of $1 \times 10^{-6} \ll 16.92$ — cases a, b, and c all pass the χ^2 test and can be said to randomly distributed random numbers.
- case d does not have a full period, but still has a χ^2 value on the order of $1 \times 10^{-6} \ll 16.92$ — case d can also be said to have randomly distributed numbers.
- Case e has a much smaller period with a period/M ratio of 2^{-12} . However, case e has a $\chi^2 = 0.005859$ which is still $\ll 16.92$ — case e still produces a random number sequence.
- Upon initial investigation, case f passes the χ^2 test with $\chi^2 < 16.92$. However, case f only has a period length of 4, which corresponds to an expected number of occurrences of 0.4 per digit. One of the criteria for a χ^2 test is the expected number of counts in any bin should not be less than one. Here, all bins have an expected count less than one, so the χ^2 test is not applicable.

2.2 Moment Test

From the moment test, we have an expected, first moment of $1/2$, which can be found via the following formula:

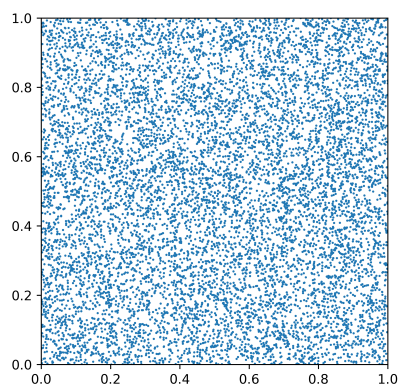
$$\mu = \frac{1}{N} \sum_{i=1}^N \approx \frac{1}{k+1} = \frac{1}{2} \quad (2)$$

In Table 2, we see cases a-e all have observed means very close to the expected mean of $1/2$. The mean residuals for these cases range from 10^{-8} to 10^{-4} , which are all at least three orders of magnitude compared to the mean on the order of 10^{-1} . However, case f had a moment of 0.258, which is roughly half of the expected moment. You cannot expect a random number generator with 4 numbers to produce a mean close to the expected mean.

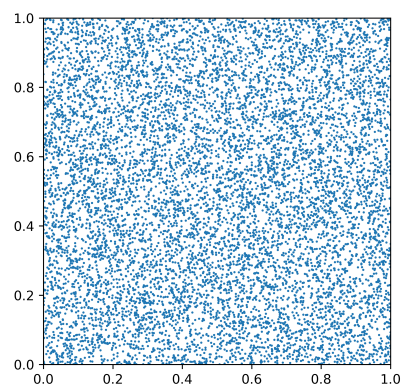
2.3 Serial Test via Plotting

Each case is plotted separately in Fig. 1. To generate these plots, I generated the entire sequence of random numbers. Then, I partitioned the odd entries into x points and the even entries into y points. In the cases where the period $> 10^4$, I sampled 10^4 random points because plotting all 2^{23} x and y combinations is intractable. In the cases where the period $< 10^4$, I plotted every point available.

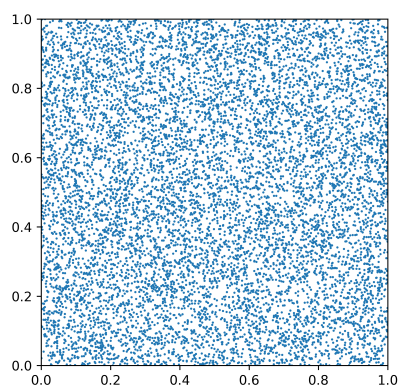
For cases a, b, and c, there is no obvious striations or groupings I can discern, which implies there is no serial correlation among the points. In case d, there is noticeable striping, so there is some serial correlation between the points. Case e has extremely obvious stripping where all the points are on three, distinct lines, which implies a very high level of serial correlation. Finally, case f has a grand total of 2 (two) points on the graph because of the period only being 4 long. From the case f plot, we can tell case f is an abject failure of a linear congruential generator.



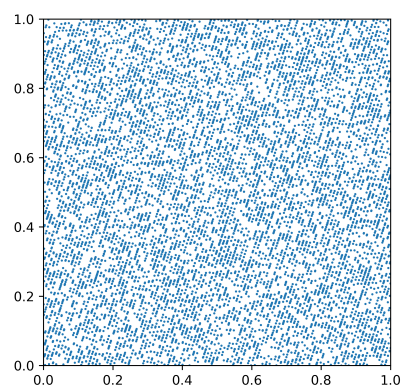
(a) Case A



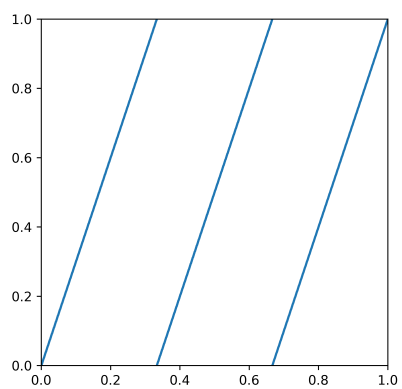
(b) Case B



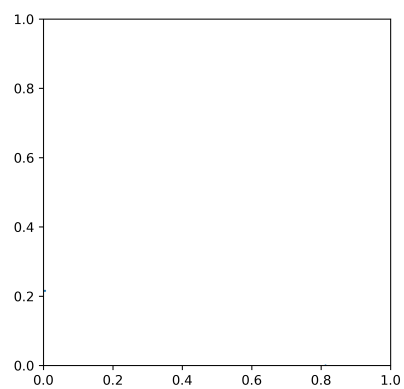
(c) Case C



(d) Case D



(e) Case E



(f) Case F

Figure 1: Serial Test via Plotting

3 Problem 3

For this problem, we first find the FFMC using the provided PDF. We do not need to normalize as we are expressly provided with a PDF. First, find the CDF:

$$P(r) = \int_0^r p(r') dr' = \int_0^r \Sigma \exp(-\Sigma r') dr' = [-\exp(-\Sigma r)]_0^r = 1 - \exp(-\Sigma r) \quad (3)$$

Next, apply the FFMC and invert to find r as a function of a random number η . Note the equivalence of $1 - \eta_1 = \eta_2$.

$$P(r) = 1 - \exp(-\Sigma r) = \eta \quad (4)$$

$$\ln(1 - \eta) = -\Sigma r \quad \Rightarrow \quad r = \frac{-\ln(\eta)}{\Sigma} \quad (5)$$

For each case, I used the full sequence of random numbers generated spanning the entire period to find the mean. For any $\eta_i < \epsilon$, I set $\eta_i = \epsilon$. Here, ϵ is a tolerance I used so the sampled r value was not infinity. From Table 3 (generated using Google Gemini), we see all cases had an MFP very close to the analytical MFP. In fact, the residual of the MFPs was roughly 6 orders of magnitude lower than the MFPs.

Table 3: Monte Carlo Mean Free Paths (MFP) ($\Sigma = 2.00 \text{ cm}^{-1}$, Analytical MFP = $\frac{1}{\Sigma} = 0.50 \text{ cm}$)

Case	a	b	x_0	Estimated MC MFP (cm)	Residual vs. Analytical (cm)
A	65539	1	1	0.500001	8.38×10^{-7}
B	65541	1	1	0.500000	2.74×10^{-7}
C	16333	1	1	0.500000	2.74×10^{-7}